



CONTINUITY

Basic Calculus

CONTINUITY OF A FUNCTION AT A NUMBER

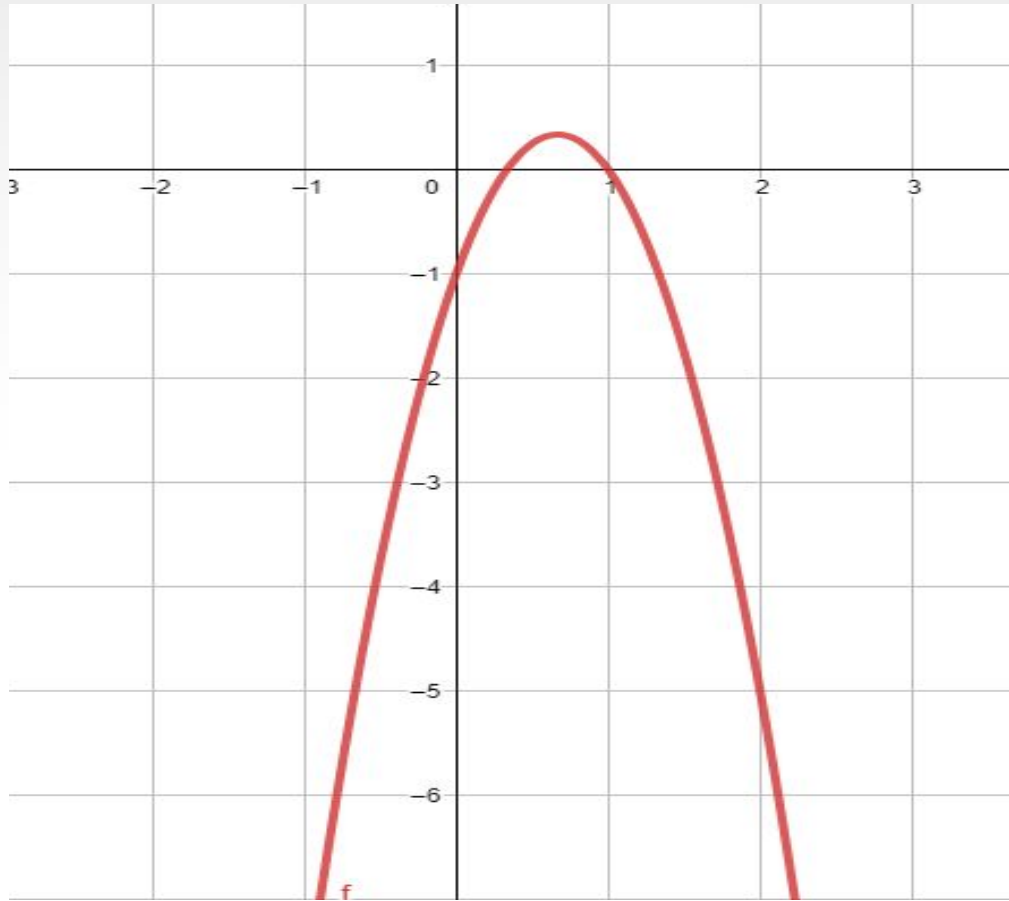
- If c is a number in the interval (a, b) and f is a function with a domain containing the interval (a, b) , then f is said to be continuous at $x = c$ if all of the following condition are satisfied:
- $f(c)$ is defined
- $\lim_{x \rightarrow c} f(x)$ exist
- $\lim_{x \rightarrow c} f(x) = f(c)$

EXAMPLE 1:

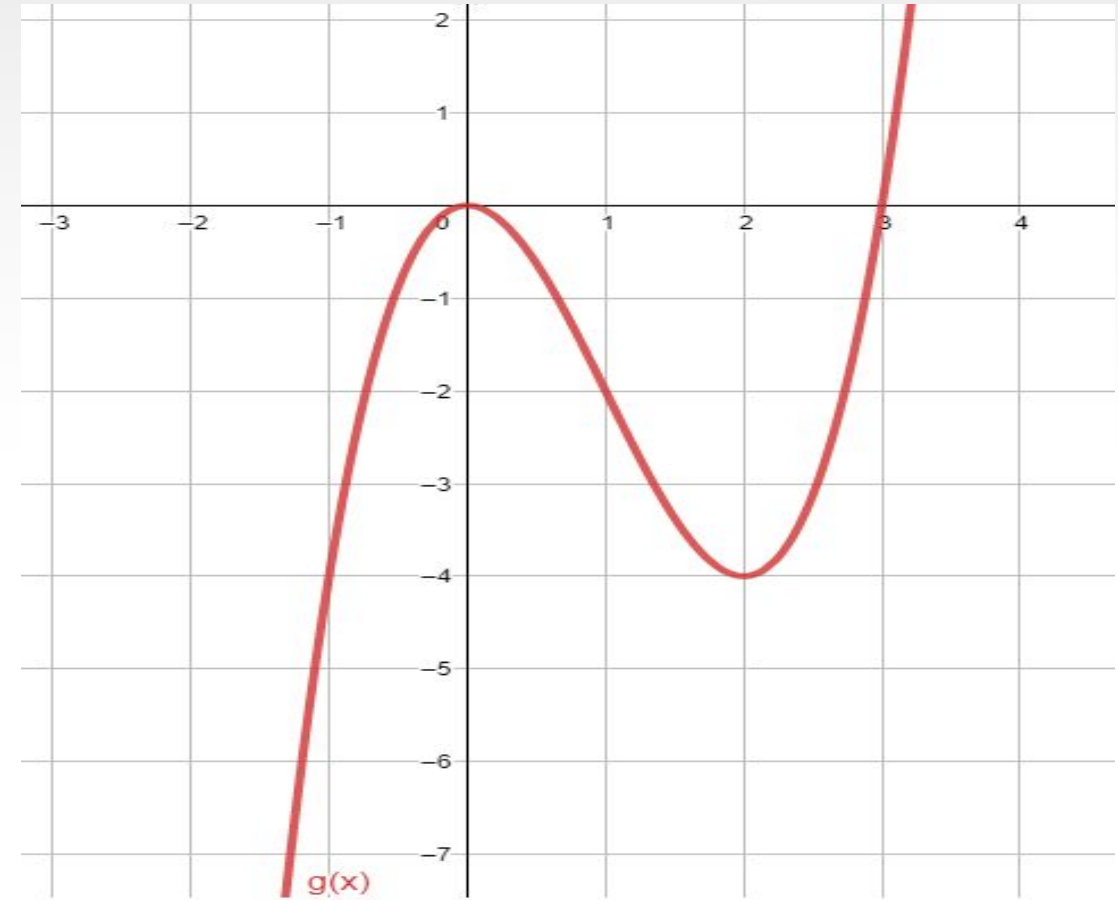
- Discuss the continuity of $f(x) = -3x^2 + 4x - 1$ and $g(x) = x^3 - 3x^2$.

SOLUTION:

- The two functions f and g are polynomial functions. Any polynomial function includes all real numbers in its domain, which implies that it is defined for every real number. Also, recall that for polynomial functions, the limit at every real number c is evaluated using direct substitution, and thus $\lim_{x \rightarrow c} f(x) = f(c)$. Therefore, the functions $f(x) = -3x^2 + 4x - 1$ and $g(x) = x^3 - 3x^2$ are continuous at every real number on the interval $(-\infty, \infty)$. Their graphs are shown in figures 6.4a and 6.4b, respectively.



(a)



(b)

Figure 6.4. Graphs of the functions (a) $f(x) = -3x^2 + 4x - 1$ and (b) $g(x) = x^3 - 3x^2$

CONTINUITY OF POLYNOMIAL FUNCTIONS

- Any polynomial functions is continuous at all real numbers.

EXAMPLE 2:

- Discuss the continuity of the function defined by $f(x) = \frac{1}{x-2}$.

SOLUTION:

- The given equation is a rational function a rational function is asymptotic at the number x that makes the denominator 0. Thus, $f(x) = \frac{1}{x-2}$ is continuous at every number in its domain, which is all real number in the intervals $(-\infty, 2)$ and $(2, \infty)$. However, the function is not continuous at $x = 2$ because $\lim_{x \rightarrow 2} \frac{1}{x-2}$ does not exist.

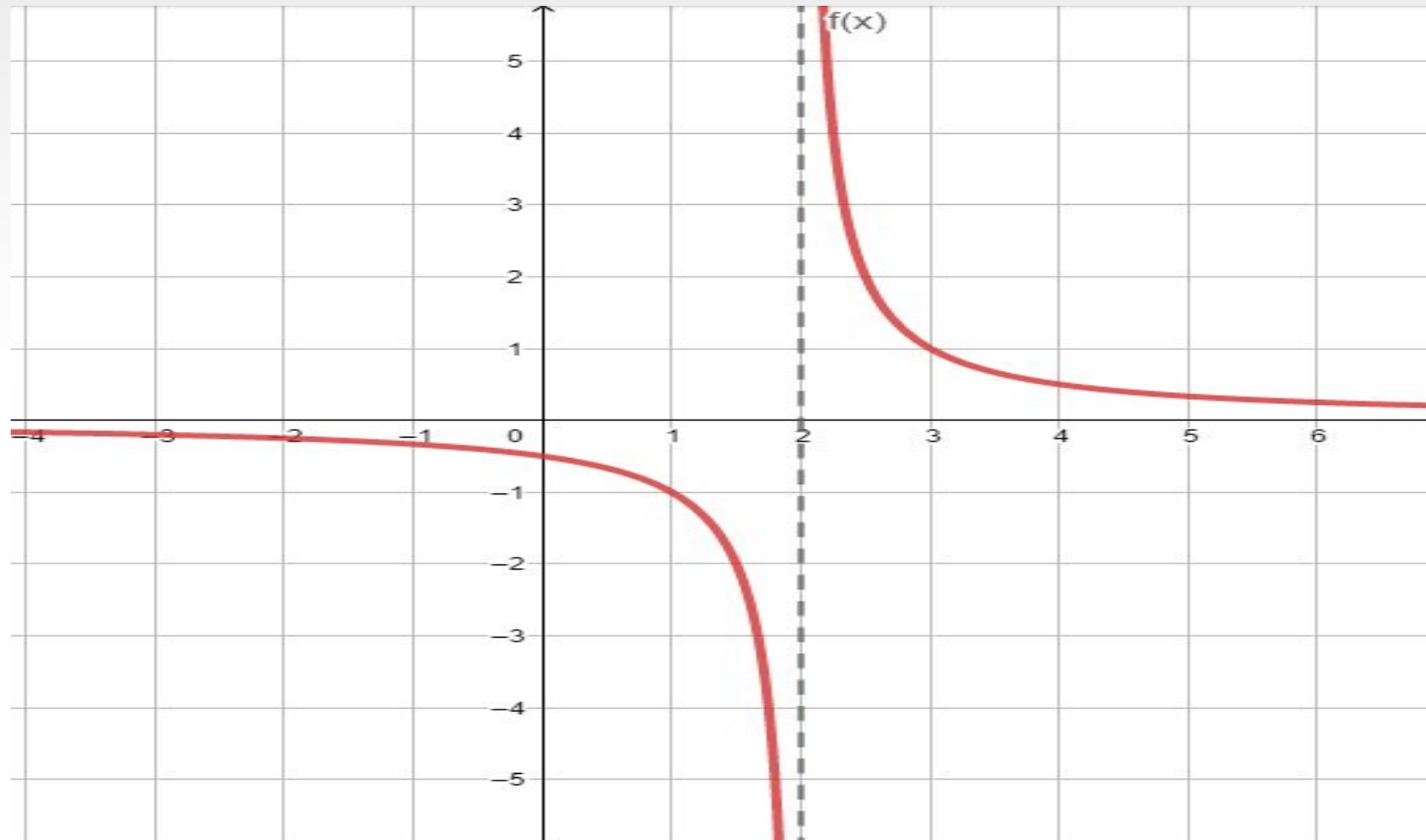


Figure 6.5

CONTINUITY OF RATIONAL FUNCTIONS

- Any rational function is continuous at any real number that makes the denominator nonzero.

EXAMPLE 3:

- Tell whether the given function is continuous at $x = 3$:

$$f(x) = \begin{cases} |x - 3|, & \text{if } x \neq 3 \\ 2, & \text{if } x = 3 \end{cases}$$

SOLUTION:

- Check if the conditions are met at $x = 3$.
- The condition $f(c)$ is defined. The function explicitly states that $f(x) = 2$ when $x = 3$. Thus, $f(3) = 2$.
- The condition $\lim_{x \rightarrow c} f(x)$ exist. By the definition of absolute value, the function can further be written as

SOLUTION:

- $$f(x) = \begin{cases} -x + 3, & \text{if } x < 3 \\ x - 3, & \text{if } x > 3 \\ 2, & x = 3 \end{cases}$$
- By definition, for the limit to exist,
 $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x).$
- $\lim_{x \rightarrow 3^-} f(x) = -3 + 3 = 0$ and $\lim_{x \rightarrow 3^+} f(x) = 3 - 3 = 0$; thus, $\lim_{x \rightarrow 3} f(x) = 0.$

SOLUTION:

- The condition $\lim_{x \rightarrow c} f(x) = f(c)$ is not satisfied because $\lim_{x \rightarrow 3} f(x) = 0$ and $f(3) = 2$. Since one of the conditions is not satisfied, the function is discontinuous at $x = 3$.

EXAMPLE 4:

- Find the discontinuities of the following function:

$$f(x) = \begin{cases} \frac{x^2 - 1}{x^2 + x}, & \text{if } x > -3 \\ 1, & \text{if } x \leq -3 \end{cases}$$

SOLUTION:

- The possible discontinuities are at $x = -1$ and $x = 0$ (the zeros of $x^2 = x$) and at $x = -3$. Because f is defined at $x = -1$ and $x = 0$, then f is not continuous at $x = -1$ and $x = 0$. Also,

$$\lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} 1 = 1$$

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \frac{x^2 - 1}{x^2 + x} = \frac{(-3)^2 - 1}{(-3)^2 - 3} = \frac{8}{6} = \frac{4}{3}$$

SOLUTION:

- Thus, $\lim_{x \rightarrow -3} f(x)$ does not exist, and therefore, f is not continuous at $x = -1$, $x = 0$, and $x = -3$.

EXAMPLE 5:

- Determine if the following function is continuous at $x = 4$;

$$f(x) = \begin{cases} -\sqrt{4 + (x - 4)^2}, & \text{if } x < 4 \\ -2, & \text{if } x = 4 \\ \frac{x - 6}{x - 3}, & \text{if } x > 4 \end{cases}$$

SOLUTION:

- $f(4) = -2$
- $\lim_{x \rightarrow 4^-} f(x) = -\sqrt{4 + (4 - 4)^2} = -\sqrt{4} = -2$
- $\lim_{x \rightarrow 4^+} f(x) = \frac{x-6}{x-3} = \frac{-2}{1} = -2$
- So, $\lim_{x \rightarrow 4} f(x) = -2$
- $f(4) = \lim_{x \rightarrow 4} f(x) = -2$
- Therefore, the function is continuous at $x = 4$.

EXAMPLE 6:

- Determine if the following function is continuous at $x = -2$:

$$g(x) = \begin{cases} 5x + 13, & \text{if } x \leq -2 \\ x^2 + 4x + 7, & \text{if } x > -2 \end{cases}$$

SOLUTION:

- $g(-2) = 5(-2) + 13 = 3$
- $\lim_{x \rightarrow -2^-} g(x) = 5(-2) + 13 = 3$
- $\lim_{x \rightarrow -2^+} g(x) = (-2)^2 + 4(-2) + 7 = 3$
- Thus, $\lim_{x \rightarrow -2} f(x) = 3$
- $g(-2) = \lim_{x \rightarrow -2} f(x) = 3$
- Therefore, the function is continuous at $x = -2$.

EXAMPLE 7:

- Find the constants A and B so that the following function is continuous at $x = 1$:

$$f(x) = \begin{cases} Bx^2 + 2Ax - 1, & \text{if } x < 1 \\ 4, & \text{if } x = 1 \\ Ax + B, & \text{if } x > 1 \end{cases}$$

SOLUTION:

- $f(1) = 4$
- $\lim_{x \rightarrow 1^-} f(x) = B(1)^2 + 2A(1) - 1 = 2A + B - 1$
- $\lim_{x \rightarrow 1^+} f(x) = A(1) + B = A + B$
- For the function to be continuous at $x = 1$, you should have $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 4$. Thus,

SOLUTION:

- $$2A + B - 1 = 4 \Leftrightarrow 2A + B = 5$$
$$A + B = 4 \Leftrightarrow B = 4 - A$$
- By substitution,
$$2A + (4 - A) = 5$$
$$2A - A = 5 - 4$$
$$A = 1$$

SOLUTION:

- To find B , substitute $A = 1$.

$$B = 4 - A$$

$$B = 4 - 1$$

$$B = 3$$

- Thus, $A = 1$ and $B = 3$.

CONTINUITY OF A FUNCTION ON A CLOSED INTERVAL

- The previous examples described the continuity of a function by determining the open interval (a, b) at which it is continuous and the real number c at which it is not continuous. This section will discuss the continuity of a function on a closed interval $[a, b]$.

CONTINUITY OF A FUNCTION ON A CLOSED INTERVAL

- If a function f is defined on a closed interval $[a, b]$ and is continuous on the open interval (a, b) , such that $\lim_{x \rightarrow a^+} f(x) = f(a)$ and $\lim_{x \rightarrow b^-} f(x) = f(b)$, then f is continuous on the closed interval $[a, b]$. Moreover, f is continuous from the right at a and continuous from the left b .

EXAMPLE 8:

- Identify the intervals wherein the function $f(x) = \sqrt{1 - x^2}$ is continuous.

SOLUTION:

- The domain of f is the set of $[-1, 1]$. Moreover,
- f is continuous from the right at $x = -1$ because

$$\lim_{x \rightarrow -1^+} \sqrt{1 - x^2} = 0 = f(-1)$$

- f is continuous from the left at $x = 1$ because

$$\lim_{x \rightarrow 1^-} \sqrt{1 - x^2} = 0 = f(1)$$

- For all intervals in which $-1 < x < 1$, the function satisfies the three conditions for continuity. Therefore, f is continuous on the closed interval $[-1, 1]$.

EXAMPLE 9:

- Identify the intervals for which the following function is continuous:

$$f(x) = \begin{cases} -x + 2, & \text{if } -1 \leq x \leq \\ \sqrt{3x - 6}, & \text{if } 2 < x \leq 5 \end{cases}$$

SOLUTION:

- The function $y = -x + 2$ is continuous on the closed interval $[-1, 2]$, and the function $y = \sqrt{3x - 6}$ is continuous on the interval $(2, 5]$. Therefore, f can be continuous on the closed interval $[-1, 5]$ as long as it is continuous at $x = 2$. To check, the three conditions of continuity at a number must be satisfied.

SOLUTION:

- $f(2) = -2 + 2 = 0$. Thus, $f(2)$ exists.
- $\lim_{x \rightarrow 2^-} f(x) = -2 + 2 = 0$ and
 $\lim_{x \rightarrow 2^+} f(x) = \sqrt{3(2) - 6} = 0$. This means that
 $\lim_{x \rightarrow 2} f(x)$ exist and is equal to 0.
- $\lim_{x \rightarrow 2} f(x) = f(2)$. Therefore, f is continuous at
 $x = 2$ and on the closed interval $[-1, 5]$.